CI/CD

Version: 1.0

Purpose

This document defines the Continuous Integration and Continuous Delivery (CI/CD) strategy of Trading Platform Pro.
The objective is to automate quality assurance, reduce integration risks and ensure a consistent software delivery process.

Objectives

The CI/CD pipeline shall provide:
• automated quality checks
• repeatable builds
• automated testing
• consistent formatting
• reliable deployments
• fast developer feedback

CI Principles

Every change should be automatically verified before integration.
The pipeline should be:
• deterministic
• reproducible
• fast
• reliable
• transparent

Continuous Integration

Every push should automatically execute:
1. Checkout Repository
2. Setup Python
3. Install Dependencies
4. Ruff Check
5. Ruff Format Verification
6. Unit Tests
7. Integration Tests (when available)
Pipeline failures must block integration until resolved.

Continuous Delivery

Future delivery stages may include:
• Build Artifacts
• Versioning
• Release Packaging
• Deployment Validation
• Release Notes
Deployment automation should remain independent from application logic.

Branch Strategy

Primary branch:
```
main
```
Optional feature branches:
```
feature/
```
Bug fixes:
```
bugfix/
```

Code Quality Gates

Before integration the following must succeed:
• Ruff
• Formatting
• Unit Tests
• Integration Tests
• Documentation validation (where applicable)

Dependencies

Development dependencies are maintained separately from production dependencies.
Examples:
• requirements.txt
• requirements-dev.txt

GitHub Actions

CI is implemented using GitHub Actions.
Typical workflow:
```
Checkout
↓
Setup Python
↓
Install Dependencies
↓
Run Ruff
↓
Run Tests
↓
Publish Results
```

Local Verification

Developers should execute before pushing:
```bash
ruff check .
ruff format .
pytest tests/unit -q
```

This reduces unnecessary CI failures.

Future Improvements

Potential enhancements include:
• coverage reporting
• security scanning
• dependency auditing
• automated releases
• container builds
• deployment automation

Release Readiness

Before a release verify:
• all quality gates passed
• documentation updated
• version numbers updated
• changelog completed
• release notes prepared
Only release software that satisfies all quality criteria.

CI/CD Review Checklist

Before enabling pipeline changes verify:
• reproducible builds
• deterministic execution
• successful test execution
• reliable quality gates
• documented workflow
• rollback strategy available

Related Documents

• Git_Workflow.md
• Testing_Strategy.md
• Development_Guidelines.md
• Runtime.md
• AGENTS.md